# PAM-4 MODEL — CHANNEL, RECEIVER AND SEPARATE LMS UPDATE PATHS

Current Python model | h0=1, h1=0.5 | CLT noise at 20 dB | 200,000 symbols | training=3,000 | reference alignment D=1

P01–P24 identify interfaces. Output names, formats and bounds appear in blocks; complete port catalogue is exported as CSV/Markdown.

## INTERFACE TYPES / RANGE ALIASES

All amplitudes and coefficient bounds are real values.
Q12 = signed 16-bit, 12 fractional bits (Q4.12).
Q = [−8, +7.999755859375]; LSB=$2^{-12}$.
S4 = {−3, −1, +1, +3}; S0 = S4 ∪ {0 at startup}.
R = [−5.999755859375, +5.99951171875] (20 dB case).
W = [−1.25,+1.25] for each FFE coefficient.
B = [−0.75,+0.75] for the DFE coefficient.
F20 = signed 40-bit state, 20 fractional bits.
μ_cfg=0.0005; μ_Q=524/$2^{20}$≈0.000499725342.
Bnd = conservative configuration bound; Obs = frame min/max.

## DELAY / STATE CONVENTIONS

ISI FIR: one internal previous-TX-symbol memory.
FFE: two internal $z^{-1}$ memories form r[n−1], r[n−2].
DFE decision $z^{-1}$: d_prev[n]=d_hat[n−1], initialized to 0.
Each LMS branch has a separate coefficient-state $z^{-1}$.
A_next[n] is computed AFTER processing symbol n.
The stored state becomes A[n+1] for the next symbol.
Q12(A[n]) supplies the visible filtering coefficients.
TX reference $z^{-D}$: aligns ground truth for training/C2.
ψ[n], x[n], d_prev[n] use the SAME iteration n.
These are model sample delays; no FPGA clock timing is implied.

## TRANSMITTER (TX)

**RANDOM PAM-4 SOURCE**
OUT P01: s[n] ∈ S4, Q12
Uniform random symbol stream
Gray labels for measurement:
−3→00; −1→01; +1→11; +3→10
Integer levels: ±4096, ±12288
200,000 symbols / frame

P01 s[n] Q12 / S4

## CHANNEL = ISI FIR + CLT NOISE + SCALE + ADDER

**ISI FIR / FIRST POSTCURSOR**
r_clean[n] = s[n] + 0.5s[n−1]
h0=1; h1=0.5
Internal $z^{-1}$ stores s[n−1] (startup 0).
OUT P02: r_clean[n], Q12
Bnd: [−4.5,+4.5]
Obs: [−4.5,+4.5]

P02 r_clean[n] Q12 / [−4.5,+4.5]

**NOISE ADDER**
Inputs: P02 r_clean, P05 η
r[n] = satQ(r_clean[n] + η[n])
OUT P06: r[n], Q12, range R
Bnd: [−5.999756,+5.999512]
Obs: [-5.4177, 5.4109]

P05 η[n] Q12 / [−1.499756,+1.499512]

**12 × TAUS88 + BIT MIX**
32-bit PRNG words per lane
Mix + extract 12-bit windows
OUT P03: (R1,...,R12)
12 × unsigned 12-bit
Each Rj ∈ [0,4095] (integer)

P03

**CLT / NORMALIZE**
S=ΣRj ∈ [0,49140]
c=S−24570
OUT P04: z_unit=c/4096
Q12; Bnd: ±5.99853515625
Mean=0; std=1

P04

**NOISE SCALE / 20 dB**
σ_float=0.2500258737
σ_Q12=1024/4096=0.25
OUT P05: η=qmul(σ_Q12,z_unit)
Q12; Bnd: [−1.499756,+1.499512]
Obs: [-0.9980, 1.0632]

## RECEIVER (RX) = FFE → DFE → SLICER → GRAY LABELS + ADAPTATION

P16 x[n] 3 × Q12 / R

TWO COEFFICIENT UPDATE BRANCHES — SAME CURRENT ERROR AND GATE

P15 ψ[n] / g[n]

**FFE LMS UPDATE**
Inputs: P16 x, P15 ψ/g, P18 A_w
Δ_w=round_F20(μ_Q ψ x)
A_w_next=P_W(A_w + Δ_w)
g=0 → hold current state
P_W projects each tap to W
OUT P17 A_w_next[n]
3 × signed40 F20 / W

P17 / F20 / W

**$z^{-1}$ / FFE COEFFICIENT STATE**
P18 A_w[n]=A_w_next[n−1] / F20 / W
P19 w[n]=Q12(A_w[n]); 3 × Q12 / W
Initial state: [0,0.75,0]

P19 w[n] / W

**DFE LMS UPDATE**
Inputs: P11 d_prev, P15 ψ/g, P21 A_b
Δ_b=round_F20(μ_Q ψ d_prev)
A_b_next=P_B(A_b − Δ_b)
g=0 → hold current state
P_B projects the tap to B
OUT P20 A_b_next[n]
signed40 F20 / B

P20 / F20 / B

**$z^{-1}$ / DFE COEFFICIENT STATE**
P21 A_b[n]=A_b_next[n−1] / F20 / B
P22 b0[n]=Q12(A_b[n]); Q12 / B
Initial state: 0

P22 b0[n] / B

P15 ψ / g

P11 / Q12 / S0

**TRAINING / DD SELECTOR**
P12 d_ref during n≤3000
P09 d_hat after n>3000
OUT P13 d_des[n]: Q12 / S0
Startup n≤D: d_des=0

P13

**C1 / LMS ERROR COMPARISON**
e_raw[n]=satQ(d_des[n]−y[n])
Inputs: P13 desired; P08 y (pre-slicer)
OUT P14 e_raw: Q12 / Q
This comparison drives adaptation.

P14

P09 d_hat[n] / S4

**SHARED ROBUST ERROR / UPDATE CONTROL**
ψ[n]=clip(e_raw[n],−2.5,+2.5); OUT P15 ψ: Q12 / [−2.5,+2.5]
g[n]=1 in training; in DD g=(|e_raw[n]|≤1.5); OUT P15 g∈{0,1}
Both LMS branches use this same ψ[n] and g[n].
No extra error-path $z^{-1}$ is inserted.

P12 aligned TX reference → training input

P08 y[n] / Q12 / Q — tap BEFORE the slicer

**FFE / 3 TAPS**
Inputs: P06 r[n], P19 w[n]
x[n]=(r[n],r[n−1],r[n−2])
Two internal $z^{-1}$ sample memories
f[n]=Σ qmul(wk,xk), k=0..2
OUT P07 f[n]: wide signed F12
Bnd: [-22.499268, 22.498535]
Obs: [-5.3218, 5.3542]
P16 x[n] to FFE LMS: 3 × Q12 / R

P06 r[n] Q12 / R

P07 f[n] / wide F12 Bnd: [−22.499268,+22.498535]

**DFE / 1 TAP + SUBTRACT**
Inputs: P07 f, P11 d_prev, P22 b0
v[n]=qmul(b0[n],d_prev[n])
Feedback v Bnd: [−2.25,+2.25]
y[n]=satQ(f[n]−v[n])
OUT P08 y[n]: Q12 / Q
Obs: [-3.9556, 3.9580]

P08 y[n] Q12 / Q

**PAM-4 SLICER**
Input P08: pre-slicer y[n]
Thresholds: −2, 0, +2
Exact threshold → lower level
OUT P09 d_hat[n]: Q12 / S4
Examples: y=0 → −1; y=2 → +1

P09 d_hat[n] Q12 / S4

**GRAY LABELS**
Symbol → Gray bits
−3/−1/+1/+3 → 00/01/11/10
OUT P10: (d_hat[n], b_hat[n])
S4 × {00,01,11,10}
Measurement postprocessing

P09 / S4

**$z^{-1}$ / DECISION MEMORY**
OUT P11: d_prev[n]=d_hat[n−1]
Q12 / S0; initialized to zero

P11 d_prev[n] Q12 / S0

**RUN SNAPSHOT / AFTER UPDATE n=200000**

| Tap | Real value | Q12 int | Hex |
|---|---|---|---|
| w0 | +0.007812500000 | 32 | 0020 |
| w1 | +0.986083984375 | 4039 | 0FC7 |
| w2 | -0.005371093750 | -22 | FFEA |
| b0 | +0.492675781250 | 2018 | 07E2 |

These coefficients are available for sample n=200001.

P10 recovered symbol + Gray bits S4 × {00,01,11,10}

P06 r[n] — raw comparison tap

## MEASUREMENT / TX REFERENCE — C0 RAW BASELINE AND C2 POST-EQ BER/SER

P01 original s[n] — no alignment delay for C0

P01 s[n] / Q12 / S4

**$z^{-D}$ / TX REFERENCE ALIGNMENT**
Input P01 s[n]; D=1
OUT P12 d_ref[n]=s[n−D]
Q12 / S0; startup reference=0
Used by training selector and C2

**C0 / RAW-RX BASELINE COMPARISON**
Inputs: P06 r[k] and P01 original s[k]
raw_decision[k]=slicer(r[k]); alignment D=0
SER: raw_decision[k] versus s[k]
BER: Gray(raw_decision[k]) XOR Gray(s[k])
Retain the same TX indices as post-EQ C2.
OUT P24: BER_raw,SER_raw ∈ [0,1]
Measured: 73,983 bit errors / 394,000 bits
BER_raw=0.18777411; SER_raw=0.37554822

**C2 / POST-EQ BER AND SER COMPARISON**
Inputs: P10 (d_hat[n],b_hat[n]); P12 d_ref[n]
SER: d_hat[n] versus d_ref[n]=s[n−D]
BER: b_hat[n] XOR Gray(d_ref[n])
Skip EQ output n≤3000 → 197,000 symbols / 394,000 bits
OUT P23: BER,SER ∈ [0,1]; integer error counters
Bit counter: 0..394,000; symbol counter: 0..197,000
Measured: BER=6/394000=1.52284×$10^{-5}$
Measured: SER=6/197000=3.04569×$10^{-5}$

P12 aligned d_ref[n]=s[n−1] / Q12 / S0

qmul: integer product → signed >>12 → sat16. FFE products are quantized separately, then summed wide. Colored dots are junctions; unmarked crossings are separate signals.

Numerical model unchanged. Two LMS branches are an explicit decomposition of its existing update equations; coefficient memories model the next-symbol state, not a delayed-error LMS variant.